

Tables by Fastener Size

Aluminum 6061-T6

$$F_{su} = 27 \text{ ksi}$$

Tables by Fastener Size

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M1.1

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	65.3	0.4	0.6

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M1.2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	72.2	0.4	0.7

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M1.4

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	83.8	0.6	0.9

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M1.6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	95.5	0.8	1.2

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M1.8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	109.4	1.0	1.6

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	121.1	1.2	1.9
1/8	242.1	2.5	3.8

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M2.2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	132.7	1.5	2.3
1/8	265.5	3.0	4.6

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M2.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	153.6	2.0	3.0
1/8	307.2	3.9	6.0

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M3

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	186.1	2.9	4.4
1/8	372.2	5.7	8.8
3/16	558.3	8.6	13.2

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M3.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	432.7	7.8	11.9
3/16	649.1	11.6	17.9

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M4

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	493.3	10.1	15.5
3/16	739.9	15.1	23.3
1/4	986.5	20.2	31.1

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M4.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	558.4	12.9	19.8
3/16	837.6	19.3	29.7
1/4	1116.8	25.7	39.6

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	623.4	16.0	24.5
3/16	935.1	23.9	36.8
1/4	1246.7	31.9	49.1
5/16	1558.4	39.9	61.4

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	744.4	22.9	35.2
3/16	1116.6	34.3	52.8
1/4	1488.9	45.7	70.3
5/16	1861.1	57.2	87.9
3/8	2233.3	68.6	105.5

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M7

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	883.6	31.7	48.7
3/16	1325.4	47.5	73.1
1/4	1767.1	63.3	97.4
5/16	2208.9	79.1	121.8
3/8	2650.7	95.0	146.1
7/16	3092.5	110.8	170.5

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	1500.3	61.4	94.5
1/4	2000.4	81.9	126.0
5/16	2500.4	102.4	157.5
3/8	3000.5	122.9	189.0
7/16	3500.6	143.3	220.5
1/2	4000.7	163.8	252.0

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M9

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	1709.0	78.7	121.1
1/4	2278.6	105.0	161.5
5/16	2848.3	131.2	201.8
3/8	3418.0	157.4	242.2
7/16	3987.6	183.7	282.6
1/2	4557.3	209.9	323.0
9/16	5126.9	236.2	363.3

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M10

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	1883.9	96.4	148.3
1/4	2511.9	128.6	197.8
5/16	3139.8	160.7	247.2
3/8	3767.8	192.8	296.7
7/16	4395.7	225.0	346.1
1/2	5023.7	257.1	395.6
9/16	5651.7	289.3	445.0
5/8	6279.6	321.4	494.5

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M11

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2092.6	117.8	181.2
1/4	2790.1	157.1	241.7
5/16	3487.7	196.4	302.1
3/8	4185.2	235.6	362.5
7/16	4882.7	274.9	422.9
1/2	5580.3	314.2	483.3
9/16	6277.8	353.4	543.7
5/8	6975.4	392.7	604.2
3/4	8370.4	471.2	725.0

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M12

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	3023.1	185.7	285.6
5/16	3778.8	232.1	357.1
3/8	4534.6	278.5	428.5
7/16	5290.4	324.9	499.9
1/2	6046.1	371.3	571.3
9/16	6801.9	417.8	642.7
5/8	7557.7	464.2	714.1
3/4	9069.2	557.0	856.9

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M14

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	3534.6	253.3	389.6
5/16	4418.2	316.6	487.0
3/8	5301.9	379.9	584.5
7/16	6185.5	443.2	681.9
1/2	7069.1	506.5	779.3
9/16	7952.8	569.8	876.7
5/8	8836.4	633.2	974.1
3/4	10603.7	759.8	1168.9
7/8	12371.0	886.4	1363.7

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M16

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	4091.2	335.0	515.4
5/16	5113.9	418.8	644.3
3/8	6136.7	502.5	773.1
7/16	7159.5	586.3	902.0
1/2	8182.3	670.0	1030.8
9/16	9205.1	753.8	1159.7
5/8	10227.9	837.6	1288.6
3/4	12273.5	1005.1	1546.3
7/8	14319.0	1172.6	1804.0
1	16364.6	1340.1	2061.7
1 1/8	18410.2	1507.6	2319.4

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M18

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	5696.6	524.8	807.4
3/8	6835.9	629.8	968.9
7/16	7975.3	734.7	1130.4
1/2	9114.6	839.7	1291.8
9/16	10253.9	944.7	1453.3
5/8	11393.2	1049.6	1614.8
3/4	13671.9	1259.5	1937.7
7/8	15950.5	1469.5	2260.7
1	18229.2	1679.4	2583.7
1 1/8	20507.8	1889.3	2906.6

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M20

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	6392.3	654.3	1006.7
3/8	7670.8	785.2	1208.0
7/16	8949.3	916.1	1409.3
1/2	10227.7	1046.9	1610.7
9/16	11506.2	1177.8	1812.0
5/8	12784.7	1308.7	2013.3
3/4	15341.6	1570.4	2416.0
7/8	17898.5	1832.1	2818.7
1	20455.5	2093.9	3221.3
1 1/8	23012.4	2355.6	3624.0
1 1/4	25569.3	2617.3	4026.7
1 3/8	28126.3	2879.1	4429.3

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M22

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	7088.1	798.1	1227.9
3/8	8505.7	957.7	1473.4
7/16	9923.3	1117.3	1719.0
1/2	11340.9	1277.0	1964.6
9/16	12758.5	1436.6	2210.1
5/8	14176.1	1596.2	2455.7
3/4	17011.4	1915.5	2946.8
7/8	19846.6	2234.7	3438.0
1	22681.8	2553.9	3929.1
1 1/8	25517.0	2873.2	4420.3
1 1/4	28352.3	3192.4	4911.4
1 3/8	31187.5	3511.7	5402.6
1 1/2	34022.7	3830.9	5893.7

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M24

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	9204.9	1130.7	1739.5
7/16	10739.0	1319.1	2029.4
1/2	12273.2	1507.6	2319.3
9/16	13807.3	1696.0	2609.3
5/8	15341.5	1884.5	2899.2
3/4	18409.8	2261.4	3479.0
7/8	21478.1	2638.2	4058.8
1	24546.4	3015.1	4638.7
1 1/8	27614.6	3392.0	5218.5
1 1/4	30682.9	3768.9	5798.4
1 3/8	33751.2	4145.8	6378.2
1 1/2	36819.5	4522.7	6958.0

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M27

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	10457.2	1445.1	2223.2
7/16	12200.1	1685.9	2593.7
1/2	13942.9	1926.8	2964.2
9/16	15685.8	2167.6	3334.8
5/8	17428.7	2408.4	3705.3
3/4	20914.4	2890.1	4446.4
7/8	24400.1	3371.8	5187.4
1	27885.8	3853.5	5928.5
1 1/8	31371.6	4335.2	6669.5
1 1/4	34857.3	4816.9	7410.6
1 3/8	38343.0	5298.6	8151.7
1 1/2	41828.8	5780.3	8892.7
1 3/4	48800.2	6743.7	10374.8

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M30

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	13503.3	2073.3	3189.8
1/2	15432.3	2369.5	3645.4
9/16	17361.4	2665.7	4101.1
5/8	19290.4	2961.9	4556.8
3/4	23148.5	3554.3	5468.1
7/8	27006.6	4146.7	6379.5
1	30864.7	4739.1	7290.9
1 1/8	34722.7	5331.4	8202.2
1 1/4	38580.8	5923.8	9113.6
1 3/8	42438.9	6516.2	10024.9
1 1/2	46297.0	7108.6	10936.3
1 3/4	54013.2	8293.4	12759.0
2	61729.3	9478.1	14581.7

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M33

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	14964.3	2527.4	3888.4
1/2	17102.1	2888.5	4443.8
9/16	19239.8	3249.6	4999.3
5/8	21377.6	3610.6	5554.8
3/4	25653.1	4332.8	6665.8
7/8	29928.6	5054.9	7776.7
1	34204.2	5777.0	8887.7
1 1/8	38479.7	6499.1	9998.7
1 1/4	42755.2	7221.3	11109.6
1 3/8	47030.7	7943.4	12220.6
1 1/2	51306.2	8665.5	13331.5
1 3/4	59857.3	10109.8	15553.5
2	68408.3	11554.0	17775.4

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M36

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	18590.9	3425.4	5269.9
9/16	20914.8	3853.6	5928.6
5/8	23238.7	4281.8	6587.3
3/4	27886.4	5138.1	7904.8
7/8	32534.1	5994.5	9222.3
1	37181.9	6850.8	10539.7
1 1/8	41829.6	7707.2	11857.2
1 1/4	46477.3	8563.5	13174.7
1 3/8	51125.1	9419.9	14492.1
1 1/2	55772.8	10276.2	15809.6
1 3/4	65068.3	11989.0	18444.5
2	74363.7	13701.7	21079.5

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M39

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	20260.7	4044.2	6221.8
9/16	22793.3	4549.7	6999.5
5/8	25325.8	5055.2	7777.2
3/4	30391.0	6066.2	9332.7
7/8	35456.2	7077.3	10888.1
1	40521.4	8088.3	12443.6
1 1/8	45586.5	9099.4	13999.0
1 1/4	50651.7	10110.4	15554.5
1 3/8	55716.9	11121.4	17109.9
1 1/2	60782.0	12132.5	18665.3
1 3/4	70912.4	14154.6	21776.2
2	81042.7	16176.6	24887.1

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M42

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	24468.2	5259.7	8091.9
5/8	27186.9	5844.1	8990.9
3/4	32624.3	7012.9	10789.1
7/8	38061.7	8181.8	12587.3
1	43499.1	9350.6	14385.5
1 1/8	48936.4	10519.4	16183.7
1 1/4	54373.8	11688.2	17981.9
1 3/8	59811.2	12857.1	19780.1
1 1/2	65248.6	14025.9	21578.3
1 3/4	76123.4	16363.5	25174.7
2	86998.1	18701.2	28771.0

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M45

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	26346.7	6068.0	9335.4
5/8	29274.1	6742.3	10372.7
3/4	35128.9	8090.7	12447.3
7/8	40983.7	9439.2	14521.8
1	46838.5	10787.6	16596.3
1 1/8	52693.4	12136.1	18670.9
1 1/4	58548.2	13484.5	20745.4
1 3/8	64403.0	14833.0	22820.0
1 1/2	70257.8	16181.4	24894.5
1 3/4	81967.5	18878.3	29043.6
2	93677.1	21575.2	33192.7

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M48

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	31135.2	7649.0	11767.6
3/4	37362.2	9178.7	14121.1
7/8	43589.2	10708.5	16474.7
1	49816.3	12238.3	18828.2
1 1/8	56043.3	13768.1	21181.7
1 1/4	62270.3	15297.9	23535.2
1 3/8	68497.4	16827.7	25888.8
1 1/2	74724.4	18357.5	28242.3
1 3/4	87178.5	21417.1	32949.3
2	99632.5	24476.7	37656.4

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M52

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	33918.1	9027.0	13887.7
3/4	40701.7	10832.4	16665.3
7/8	47485.3	12637.8	19442.8
1	54268.9	14443.2	22220.3
1 1/8	61052.5	16248.6	24997.9
1 1/4	67836.1	18054.0	27775.4
1 3/8	74619.8	19859.4	30553.0
1 1/2	81403.4	21664.8	33330.5
1 3/4	94970.6	25275.6	38885.6
2	108537.8	28886.4	44440.7

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M56

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	43770.7	12545.3	19300.5
7/8	51065.8	14636.2	22517.2
1	58360.9	16727.1	25733.9
1 1/8	65656.0	18817.9	28950.7
1 1/4	72951.1	20908.8	32167.4
1 3/8	80246.2	22999.7	35384.2
1 1/2	87541.3	25090.6	38600.9
1 3/4	102131.6	29272.4	45034.4
2	116721.8	33454.1	51467.9

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M60

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	47110.2	14466.9	22256.8
7/8	54961.9	16878.0	25966.2
1	62813.5	19289.2	29675.7
1 1/8	70665.2	21700.3	33385.2
1 1/4	78516.9	24111.5	37094.6
1 3/8	86368.6	26522.6	40804.1
1 1/2	94220.3	28933.8	44513.5
1 3/4	109923.7	33756.1	51932.5
2	125627.1	38578.4	59351.4

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M64

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	50178.3	16436.4	25286.7
7/8	58541.4	19175.8	29501.2
1	66904.4	21915.1	33715.6
1 1/8	75267.5	24654.5	37930.1
1 1/4	83630.5	27393.9	42144.5
1 3/8	91993.6	30133.3	46359.0
1 1/2	100356.6	32872.7	50573.4
1 3/4	117082.7	38351.5	59002.3
2	133808.8	43830.3	67431.2

Aluminum 6061-T6 ($F_{su} = 27$ ksi, $K = 0.20$)

Fastener Size = M68

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	53517.8	18625.9	28655.2
7/8	62437.4	21730.2	33431.1
1	71357.1	24834.5	38206.9
1 1/8	80276.7	27938.8	42982.8
1 1/4	89196.3	31043.1	47758.7
1 3/8	98116.0	34147.4	52534.5
1 1/2	107035.6	37251.8	57310.4
1 3/4	124874.9	43460.4	66862.1
2	142714.1	49669.0	76413.9

AISI 304 Stainless Steel

$$F_{su} = 50 \text{ ksi}$$

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	224.2	2.3	3.5

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M2.2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	245.8	2.8	4.3

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M2.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	284.5	3.6	5.6

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M3

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	344.6	5.3	8.1

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M4

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	913.5	18.7	28.8

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M4.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1034.1	23.8	36.6

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1154.4	29.5	45.4

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1378.6	42.3	65.1
3/16	2067.9	63.5	97.7

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M7

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1636.2	58.6	90.2
3/16	2454.4	87.9	135.3
1/4	3272.5	117.2	180.4

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2778.3	113.8	175.0
1/4	3704.4	151.7	233.3

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M9

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	3164.8	145.8	224.3
1/4	4219.7	194.4	299.0
5/16	5274.6	243.0	373.8

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M10

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	3488.7	178.6	274.7
1/4	4651.6	238.1	366.3
5/16	5814.5	297.6	457.8

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M11

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	3875.2	218.2	335.6
1/4	5166.9	290.9	447.5
5/16	6458.7	363.6	559.4
3/8	7750.4	436.3	671.3

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M12

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	5598.3	343.8	529.0
5/16	6997.9	429.8	661.2
3/8	8397.4	515.7	793.5
7/16	9797.0	601.7	925.7

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M14

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	6545.5	469.0	721.6
5/16	8181.9	586.3	901.9
3/8	9818.3	703.5	1082.3
7/16	11454.6	820.8	1262.7
1/2	13091.0	938.0	1443.1

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M16

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	7576.2	620.4	954.5
5/16	9470.3	775.5	1193.1
3/8	11364.3	930.6	1431.7
7/16	13258.4	1085.7	1670.3
1/2	15152.4	1240.8	1909.0
9/16	17046.5	1395.9	2147.6

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M18

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	10549.3	971.9	1495.2
3/8	12659.1	1166.2	1794.2
7/16	14769.0	1360.6	2093.2
1/2	16878.8	1555.0	2392.3
9/16	18988.7	1749.4	2691.3
5/8	21098.6	1943.7	2990.3

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M20

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	11837.7	1211.7	1864.2
3/8	14205.2	1454.1	2237.0
7/16	16572.7	1696.4	2609.9
1/2	18940.3	1938.8	2982.7
9/16	21307.8	2181.1	3355.6
5/8	23675.3	2423.5	3728.4
3/4	28410.4	2908.1	4474.1

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M22

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	13126.0	1478.0	2273.8
3/8	15751.3	1773.6	2728.6
7/16	18376.5	2069.2	3183.3
1/2	21001.7	2364.8	3638.1
9/16	23626.9	2660.3	4092.8
5/8	26252.1	2955.9	4547.6
3/4	31502.5	3547.1	5457.1

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M24

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	17046.1	2093.8	3221.3
7/16	19887.1	2442.8	3758.2
1/2	22728.1	2791.8	4295.1
9/16	25569.1	3140.8	4832.0
5/8	28410.1	3489.7	5368.8
3/4	34092.2	4187.7	6442.6
7/8	39774.2	4885.6	7516.4

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M27

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	19365.2	2676.1	4117.0
7/16	22592.7	3122.1	4803.2
1/2	25820.2	3568.1	5489.3
9/16	29047.8	4014.1	6175.5
5/8	32275.3	4460.1	6861.7
3/4	38730.3	5352.1	8234.0
7/8	45185.4	6244.1	9606.3
1	51640.4	7136.1	10978.7

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M30

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	25006.1	3839.5	5907.0
1/2	28578.4	4388.0	6750.8
9/16	32150.7	4936.5	7594.7
5/8	35723.0	5485.0	8438.5
3/4	42867.6	6582.0	10126.2
7/8	50012.2	7679.0	11813.9
1	57156.8	8776.0	13501.6
1 1/8	64301.4	9873.0	15189.3

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M33

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	27711.7	4680.4	7200.7
1/2	31670.5	5349.1	8229.3
9/16	35629.3	6017.7	9258.0
5/8	39588.1	6686.3	10286.7
3/4	47505.8	8023.6	12344.0
7/8	55423.4	9360.9	14401.4
1	63341.0	10698.1	16458.7
1 1/8	71258.6	12035.4	18516.0
1 1/4	79176.3	13372.7	20573.4

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M36****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	34427.7	6343.4	9759.0
9/16	38731.1	7136.3	10978.9
5/8	43034.6	7929.2	12198.8
3/4	51641.5	9515.0	14638.5
7/8	60248.4	11100.9	17078.3
1	68855.3	12686.7	19518.0
1 1/8	77462.2	14272.6	21957.8
1 1/4	86069.1	15858.4	24397.5
1 3/8	94676.0	17444.2	26837.3

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M39****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	37519.8	7489.2	11521.8
9/16	42209.7	8425.3	12962.0
5/8	46899.7	9361.5	14402.3
3/4	56279.7	11233.8	17282.7
7/8	65659.6	13106.1	20163.2
1	75039.5	14978.4	23043.6
1 1/8	84419.5	16850.7	25924.1
1 1/4	93799.4	18723.0	28804.5
1 3/8	103179.4	20595.3	31685.0
1 1/2	112559.3	22467.5	34565.5

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M42****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	45311.5	9740.2	14984.9
5/8	50346.1	10822.4	16649.9
3/4	60415.4	12986.9	19979.9
7/8	70484.6	15151.4	23309.9
1	80553.8	17315.9	26639.8
1 1/8	90623.0	19480.4	29969.8
1 1/4	100692.3	21644.9	33299.8
1 3/8	110761.5	23809.4	36629.8
1 1/2	120830.7	25973.8	39959.8

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M45****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	48790.2	11237.1	17287.9
5/8	54211.3	12485.7	19208.7
3/4	65053.5	14982.8	23050.5
7/8	75895.8	17479.9	26892.2
1	86738.1	19977.1	30734.0
1 1/8	97580.3	22474.2	34575.7
1 1/4	108422.6	24971.3	38417.4
1 3/8	119264.8	27468.5	42259.2
1 1/2	130107.1	29965.6	46100.9
1 3/4	151791.6	34959.9	53784.4

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M48****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	57657.7	14164.7	21791.9
3/4	69189.2	16997.7	26150.3
7/8	80720.8	19830.6	30508.6
1	92252.3	22663.6	34867.0
1 1/8	103783.9	25496.5	39225.4
1 1/4	115315.4	28329.5	43583.8
1 3/8	126847.0	31162.4	47942.2
1 1/2	138378.5	33995.3	52300.5
1 3/4	161441.6	39661.2	61017.3

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M52****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	62811.2	16716.7	25718.0
3/4	75373.5	20060.0	30861.6
7/8	87935.7	23403.4	36005.2
1	100498.0	26746.7	41148.8
1 1/8	113060.2	30090.0	46292.4
1 1/4	125622.5	33433.4	51436.0
1 3/8	138184.7	36776.7	56579.6
1 1/2	150747.0	40120.1	61723.2
1 3/4	175871.5	46806.7	72010.4
2	200996.0	53493.4	82297.6

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M56****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	81056.8	23232.0	35741.6
7/8	94566.3	27104.0	41698.5
1	108075.7	30976.0	47655.4
1 1/8	121585.2	34848.0	53612.4
1 1/4	135094.7	38720.0	59569.3
1 3/8	148604.1	42592.1	65526.2
1 1/2	162113.6	46464.1	71483.2
1 3/4	189132.5	54208.1	83397.0
2	216151.5	61952.1	95310.9

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M60****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	87241.0	26790.6	41216.2
7/8	101781.2	31255.6	48085.6
1	116321.4	35720.7	54955.0
1 1/8	130861.6	40185.8	61824.4
1 1/4	145401.7	44650.9	68693.7
1 3/8	159941.9	49116.0	75563.1
1 1/2	174482.1	53581.1	82432.5
1 3/4	203562.4	62511.3	96171.2
2	232642.8	71441.5	109910.0

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)**Fastener Size = M64****[NOT TO BE USED FOR FLIGHT HARDWARE]**

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	92922.8	30437.7	46827.2
7/8	108409.9	35510.7	54631.8
1	123897.1	40583.6	62436.3
1 1/8	139384.2	45656.6	70240.9
1 1/4	154871.3	50729.5	78045.4
1 3/8	170358.5	55802.5	85849.9
1 1/2	185845.6	60875.4	93654.5
1 3/4	216819.9	71021.3	109263.6
2	247794.2	81167.2	124872.6

304 Stainless Steel ($F_{su} = 50$ ksi, $K = 0.20$)

Fastener Size = M68

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	99107.0	34492.4	53065.2
7/8	115624.9	40241.1	61909.4
1	132142.7	45989.8	70753.6
1 1/8	148660.6	51738.6	79597.8
1 1/4	165178.4	57487.3	88442.0
1 3/8	181696.2	63236.0	97286.2
1 1/2	198214.1	68984.7	106130.4
1 3/4	231249.8	80482.2	123818.8
2	264285.5	91979.7	141507.2

ASTM A36 Steel

$$F_{su} = 34.8 \text{ ksi}$$

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M1.4

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	108.1	0.8	1.2

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M1.6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	123.1	1.0	1.6

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M1.8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	141.1	1.3	2.0

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	156.0	1.6	2.5

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M2.2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	171.1	1.9	3.0

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M2.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	198.0	2.5	3.9
1/8	396.0	5.1	7.8

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M3

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	239.9	3.7	5.7
1/8	479.7	7.4	11.3

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M3.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	557.8	10.0	15.4

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M4

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	635.8	13.0	20.0
3/16	953.7	19.5	30.0

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M4.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	719.7	16.6	25.5
3/16	1079.6	24.9	38.3

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	803.5	20.6	31.6
3/16	1205.2	30.8	47.4
1/4	1606.9	41.1	63.3

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	959.5	29.5	45.3
3/16	1439.2	44.2	68.0
1/4	1919.0	58.9	90.7

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M7

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1138.8	40.8	62.8
3/16	1708.2	61.2	94.2
1/4	2277.7	81.6	125.5
5/16	2847.1	102.0	156.9

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	1933.7	79.2	121.8
1/4	2578.2	105.6	162.4
5/16	3222.8	132.0	203.0
3/8	3867.4	158.3	243.6

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M9

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2202.7	101.5	156.1
1/4	2936.9	135.3	208.1
5/16	3671.1	169.1	260.2
3/8	4405.4	202.9	312.2
7/16	5139.6	236.7	364.2

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M10

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2428.1	124.3	191.2
1/4	3237.5	165.7	254.9
5/16	4046.9	207.1	318.7
3/8	4856.2	248.5	382.4
7/16	5665.6	290.0	446.1
1/2	6475.0	331.4	509.8

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M11

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2697.1	151.8	233.6
1/4	3596.2	202.5	311.5
5/16	4495.2	253.1	389.4
3/8	5394.3	303.7	467.2
7/16	6293.3	354.3	545.1
1/2	7192.4	404.9	623.0
9/16	8091.4	455.5	700.8

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M12

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	3896.4	239.3	368.2
5/16	4870.5	299.1	460.2
3/8	5844.6	359.0	552.2
7/16	6818.7	418.8	644.3
1/2	7792.8	478.6	736.3
9/16	8766.9	538.4	828.4
5/8	9741.0	598.3	920.4

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M14

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	4555.7	326.4	502.2
5/16	5694.6	408.0	627.7
3/8	6833.5	489.6	753.3
7/16	7972.4	571.3	878.8
1/2	9111.3	652.9	1004.4
9/16	10250.3	734.5	1129.9
5/8	11389.2	816.1	1255.5

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M16

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	5273.0	431.8	664.3
5/16	6591.3	539.8	830.4
3/8	7909.6	647.7	996.5
7/16	9227.8	755.7	1162.6
1/2	10546.1	863.6	1328.6
9/16	11864.3	971.6	1494.7
5/8	13182.6	1079.5	1660.8
3/4	15819.1	1295.4	1993.0
7/8	18455.6	1511.3	2325.1

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M18

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	7342.3	676.4	1040.6
3/8	8810.8	811.7	1248.8
7/16	10279.2	947.0	1456.9
1/2	11747.7	1082.3	1665.0
9/16	13216.1	1217.5	1873.2
5/8	14684.6	1352.8	2081.3
3/4	17621.5	1623.4	2497.5
7/8	20558.4	1894.0	2913.8

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M20

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	8239.0	843.4	1297.5
3/8	9886.8	1012.0	1557.0
7/16	11534.6	1180.7	1816.5
1/2	13182.4	1349.4	2076.0
9/16	14830.2	1518.1	2335.5
5/8	16478.0	1686.7	2595.0
3/4	19773.6	2024.1	3114.0
7/8	23069.2	2361.4	3633.0
1	26364.8	2698.8	4151.9

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M22

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	9135.7	1028.7	1582.6
3/8	10962.9	1234.4	1899.1
7/16	12790.0	1440.1	2215.6
1/2	14617.2	1645.9	2532.1
9/16	16444.3	1851.6	2848.6
5/8	18271.5	2057.3	3165.1
3/4	21925.7	2468.8	3798.2
7/8	25580.0	2880.3	4431.2
1	29234.3	3291.7	5064.2
1 1/8	32888.6	3703.2	5697.2

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M24

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	11864.1	1457.3	2242.0
7/16	13841.4	1700.2	2615.7
1/2	15818.8	1943.1	2989.4
9/16	17796.1	2186.0	3363.0
5/8	19773.5	2428.9	3736.7
3/4	23728.1	2914.6	4484.1
7/8	27682.8	3400.4	5231.4
1	31637.5	3886.2	5978.7
1 1/8	35592.2	4372.0	6726.1
1 1/4	39546.9	4857.7	7473.4

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M27

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	13478.2	1862.5	2865.4
7/16	15724.5	2173.0	3343.0
1/2	17970.9	2483.4	3820.6
9/16	20217.2	2793.8	4298.2
5/8	22463.6	3104.2	4775.7
3/4	26956.3	3725.1	5730.9
7/8	31449.0	4345.9	6686.0
1	35941.7	4966.8	7641.2
1 1/8	40434.5	5587.6	8596.3
1 1/4	44927.2	6208.4	9551.4
1 3/8	49419.9	6829.3	10506.6
1 1/2	53912.6	7450.1	11461.7

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M30

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	17404.2	2672.3	4111.2
1/2	19890.6	3054.1	4698.6
9/16	22376.9	3435.8	5285.9
5/8	24863.2	3817.6	5873.2
3/4	29835.8	4581.1	7047.8
7/8	34808.5	5344.6	8222.5
1	39781.1	6108.1	9397.1
1 1/8	44753.8	6871.6	10571.8
1 1/4	49726.4	7635.2	11746.4
1 3/8	54699.0	8398.7	12921.0
1 1/2	59671.7	9162.2	14095.7

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M33

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	19287.3	3257.6	5011.7
1/2	22042.7	3723.0	5727.6
9/16	24798.0	4188.3	6443.6
5/8	27553.3	4653.7	7159.5
3/4	33064.0	5584.4	8591.4
7/8	38574.7	6515.2	10023.3
1	44085.4	7445.9	11455.2
1 1/8	49596.0	8376.7	12887.2
1 1/4	55106.7	9307.4	14319.1
1 3/8	60617.4	10238.1	15751.0
1 1/2	66128.0	11168.9	17182.9
1 3/4	77149.4	13030.3	20046.7

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M36

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	23961.6	4415.0	6792.3
9/16	26956.9	4966.9	7641.3
5/8	29952.1	5518.7	8490.3
3/4	35942.5	6622.5	10188.4
7/8	41932.9	7726.2	11886.5
1	47923.3	8830.0	13584.6
1 1/8	53913.7	9933.7	15282.6
1 1/4	59904.1	11037.5	16980.7
1 3/8	65894.5	12141.2	18678.8
1 1/2	71884.9	13244.9	20376.8
1 3/4	83865.8	15452.4	23773.0
2	95846.6	17659.9	27169.1

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M39

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	26113.8	5212.5	8019.2
9/16	29378.0	5864.0	9021.6
5/8	32642.2	6515.6	10024.0
3/4	39170.6	7818.7	12028.8
7/8	45699.1	9121.8	14033.6
1	52227.5	10424.9	16038.4
1 1/8	58756.0	11728.1	18043.2
1 1/4	65284.4	13031.2	20048.0
1 3/8	71812.8	14334.3	22052.8
1 1/2	78341.3	15637.4	24057.6
1 3/4	91398.2	18243.6	28067.2
2	104455.0	20849.9	32076.7

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M42

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	31536.8	6779.2	10429.5
5/8	35040.9	7532.4	11588.3
3/4	42049.1	9038.9	13906.0
7/8	49057.3	10545.4	16223.7
1	56065.5	12051.9	18541.3
1 1/8	63073.6	13558.3	20859.0
1 1/4	70081.8	15064.8	23176.7
1 3/8	77090.0	16571.3	25494.3
1 1/2	84098.2	18077.8	27812.0
1 3/4	98114.5	21090.8	32447.3
2	112130.9	24103.7	37082.7

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M45

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	33957.9	7821.0	12032.3
5/8	37731.1	8690.0	13369.3
3/4	45277.3	10428.0	16043.1
7/8	52823.5	12166.0	18717.0
1	60369.7	13904.0	21390.8
1 1/8	67915.9	15642.0	24064.7
1 1/4	75462.1	17380.1	26738.5
1 3/8	83008.3	19118.1	29412.4
1 1/2	90554.5	20856.1	32086.3
1 3/4	105646.9	24332.1	37434.0
2	120739.4	27808.1	42781.7

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M48

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	40129.8	9858.7	15167.2
3/4	48155.7	11830.4	18200.6
7/8	56181.7	13802.1	21234.0
1	64207.6	15773.8	24267.4
1 1/8	72233.6	17745.6	27300.9
1 1/4	80259.5	19717.3	30334.3
1 3/8	88285.5	21689.0	33367.7
1 1/2	96311.4	23660.8	36401.2
1 3/4	112363.3	27604.2	42468.0
2	128415.2	31547.7	48534.9

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M52

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	43716.6	11634.8	17899.7
3/4	52459.9	13961.8	21479.7
7/8	61203.3	16288.7	25059.6
1	69946.6	18615.7	28639.6
1 1/8	78689.9	20942.7	32219.5
1 1/4	87433.2	23269.6	35799.4
1 3/8	96176.6	25596.6	39379.4
1 1/2	104919.9	27923.6	42959.3
1 3/4	122406.5	32577.5	50119.2
2	139893.2	37231.4	57279.1

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M56

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	56415.5	16169.5	24876.1
7/8	65818.1	18864.4	29022.2
1	75220.7	21559.3	33168.2
1 1/8	84623.3	24254.2	37314.2
1 1/4	94025.9	26949.2	41460.2
1 3/8	103428.5	29644.1	45606.3
1 1/2	112831.1	32339.0	49752.3
1 3/4	131636.2	37728.8	58044.3
2	150441.4	43118.6	66336.4

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M60

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	60719.8	18646.2	28686.5
7/8	70839.7	21753.9	33467.6
1	80959.7	24861.6	38248.7
1 1/8	91079.6	27969.3	43029.8
1 1/4	101199.6	31077.0	47810.8
1 3/8	111319.6	34184.7	52591.9
1 1/2	121439.5	37292.5	57373.0
1 3/4	141679.4	43507.9	66935.2
2	161919.4	49723.3	76497.3

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M64

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	64674.3	21184.6	32591.8
7/8	75453.3	24715.4	38023.7
1	86232.4	28246.2	43455.7
1 1/8	97011.4	31777.0	48887.6
1 1/4	107790.5	35307.7	54319.6
1 3/8	118569.5	38838.5	59751.6
1 1/2	129348.5	42369.3	65183.5
1 3/4	150906.6	49430.8	76047.4
2	172464.7	56492.4	86911.4

ASTM A36 Steel ($F_{su} = 34.8$ ksi, $K = 0.20$)

Fastener Size = M68

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	68978.5	24006.7	36933.4
7/8	80474.9	28007.8	43088.9
1	91971.3	32008.9	49244.5
1 1/8	103467.8	36010.0	55400.1
1 1/4	114964.2	40011.2	61555.6
1 3/8	126460.6	44012.3	67711.2
1 1/2	137957.0	48013.4	73866.7
1 3/4	160949.8	56015.6	86177.9
2	183942.7	64017.8	98489.0

Aluminum 2024-T4/T351

$$F_{su} = 37.5 \text{ ksi}$$

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M1.6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	132.7	1.1	1.7

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M1.8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	152.0	1.4	2.2

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	168.1	1.7	2.6

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M2.2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	184.4	2.1	3.2

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M2.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	213.4	2.7	4.2

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M3

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	258.5	4.0	6.1
1/8	517.0	7.9	12.2

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M3.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	601.0	10.8	16.6

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M4

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	685.1	14.0	21.6
3/16	1027.6	21.0	32.4

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M4.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	775.5	17.9	27.5
3/16	1163.3	26.8	41.2

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	865.8	22.2	34.1
3/16	1298.7	33.2	51.1

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1033.9	31.8	48.8
3/16	1550.9	47.6	73.3
1/4	2067.9	63.5	97.7

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M7

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1227.2	44.0	67.6
3/16	1840.8	65.9	101.5
1/4	2454.4	87.9	135.3
5/16	3068.0	109.9	169.1

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2083.7	85.3	131.3
1/4	2778.3	113.8	175.0
5/16	3472.8	142.2	218.8
3/8	4167.4	170.6	262.5

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M9

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2373.6	109.3	168.2
1/4	3164.8	145.8	224.3
5/16	3956.0	182.2	280.3
3/8	4747.2	218.7	336.4
7/16	5538.4	255.1	392.5

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M10

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2616.5	133.9	206.0
1/4	3488.7	178.6	274.7
5/16	4360.9	223.2	343.4
3/8	5233.0	267.8	412.0
7/16	6105.2	312.5	480.7

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M11

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2906.4	163.6	251.7
1/4	3875.2	218.2	335.6
5/16	4844.0	272.7	419.6
3/8	5812.8	327.3	503.5
7/16	6781.6	381.8	587.4
1/2	7750.4	436.3	671.3

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M12

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	4198.7	257.9	396.7
5/16	5248.4	322.3	495.9
3/8	6298.1	386.8	595.1
7/16	7347.7	451.3	694.3
1/2	8397.4	515.7	793.5
9/16	9447.1	580.2	892.6

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M14

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	4909.1	351.8	541.2
5/16	6136.4	439.7	676.5
3/8	7363.7	527.6	811.7
7/16	8591.0	615.6	947.0
1/2	9818.3	703.5	1082.3
9/16	11045.5	791.5	1217.6
5/8	12272.8	879.4	1352.9

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M16

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	5682.2	465.3	715.9
5/16	7102.7	581.6	894.8
3/8	8523.2	698.0	1073.8
7/16	9943.8	814.3	1252.8
1/2	11364.3	930.6	1431.7
9/16	12784.8	1046.9	1610.7
5/8	14205.4	1163.3	1789.7
3/4	17046.5	1395.9	2147.6

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M18

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	7912.0	728.9	1121.4
3/8	9494.4	874.7	1345.7
7/16	11076.7	1020.5	1569.9
1/2	12659.1	1166.2	1794.2
9/16	14241.5	1312.0	2018.5
5/8	15823.9	1457.8	2242.8
3/4	18988.7	1749.4	2691.3
7/8	22153.5	2040.9	3139.9

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M20

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	8878.2	908.8	1398.1
3/8	10653.9	1090.6	1677.8
7/16	12429.5	1272.3	1957.4
1/2	14205.2	1454.1	2237.0
9/16	15980.8	1635.8	2516.7
5/8	17756.5	1817.6	2796.3
3/4	21307.8	2181.1	3355.6
7/8	24859.1	2544.6	3914.8
1	28410.4	2908.1	4474.1

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M22

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	9844.5	1108.5	1705.4
3/8	11813.4	1330.2	2046.4
7/16	13782.3	1551.9	2387.5
1/2	15751.3	1773.6	2728.6
9/16	17720.2	1995.3	3069.6
5/8	19689.1	2217.0	3410.7
3/4	23626.9	2660.3	4092.8
7/8	27564.7	3103.7	4775.0
1	31502.5	3547.1	5457.1
1 1/8	35440.3	3990.5	6139.3

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M24

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	12784.6	1570.4	2416.0
7/16	14915.3	1832.1	2818.6
1/2	17046.1	2093.8	3221.3
9/16	19176.8	2355.6	3624.0
5/8	21307.6	2617.3	4026.6
3/4	25569.1	3140.8	4832.0
7/8	29830.6	3664.2	5637.3
1	34092.2	4187.7	6442.6
1 1/8	38353.7	4711.2	7247.9

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M27

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	14523.9	2007.0	3087.8
7/16	16944.5	2341.5	3602.4
1/2	19365.2	2676.1	4117.0
9/16	21785.8	3010.6	4631.6
5/8	24206.5	3345.1	5146.3
3/4	29047.8	4014.1	6175.5
7/8	33889.0	4683.1	7204.8
1	38730.3	5352.1	8234.0
1 1/8	43571.6	6021.1	9263.3
1 1/4	48412.9	6690.1	10292.5
1 3/8	53254.2	7359.1	11321.8

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M30

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	18754.6	2879.6	4430.2
1/2	21433.8	3291.0	5063.1
9/16	24113.0	3702.4	5696.0
5/8	26792.2	4113.8	6328.9
3/4	32150.7	4936.5	7594.7
7/8	37509.1	5759.3	8860.4
1	42867.6	6582.0	10126.2
1 1/8	48226.0	7404.8	11392.0
1 1/4	53584.5	8227.5	12657.8
1 3/8	58942.9	9050.3	13923.5
1 1/2	64301.4	9873.0	15189.3

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M33

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	20783.8	3510.3	5400.5
1/2	23752.9	4011.8	6172.0
9/16	26722.0	4513.3	6943.5
5/8	29691.1	5014.8	7715.0
3/4	35629.3	6017.7	9258.0
7/8	41567.5	7020.7	10801.0
1	47505.8	8023.6	12344.0
1 1/8	53444.0	9026.6	13887.0
1 1/4	59382.2	10029.5	15430.0
1 3/8	65320.4	11032.5	16973.0
1 1/2	71258.6	12035.4	18516.0

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M36

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	25820.7	4757.5	7319.3
9/16	29048.3	5352.2	8234.2
5/8	32275.9	5946.9	9149.1
3/4	38731.1	7136.3	10978.9
7/8	45186.3	8325.7	12808.7
1	51641.5	9515.0	14638.5
1 1/8	58096.7	10704.4	16468.3
1 1/4	64551.8	11893.8	18298.2
1 3/8	71007.0	13083.2	20128.0
1 1/2	77462.2	14272.6	21957.8
1 3/4	90372.6	16651.3	25617.4

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M39

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	28139.8	5616.9	8641.4
9/16	31657.3	6319.0	9721.5
5/8	35174.8	7021.1	10801.7
3/4	42209.7	8425.3	12962.0
7/8	49244.7	9829.6	15122.4
1	56279.7	11233.8	17282.7
1 1/8	63314.6	12638.0	19443.1
1 1/4	70349.6	14042.2	21603.4
1 3/8	77384.5	15446.4	23763.8
1 1/2	84419.5	16850.7	25924.1
1 3/4	98489.4	19659.1	30244.8
2	112559.3	22467.5	34565.5

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M42

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	33983.6	7305.1	11238.7
5/8	37759.6	8116.8	12487.4
3/4	45311.5	9740.2	14984.9
7/8	52863.4	11363.6	17482.4
1	60415.4	12986.9	19979.9
1 1/8	67967.3	14610.3	22477.4
1 1/4	75519.2	16233.7	24974.9
1 3/8	83071.1	17857.0	27472.3
1 1/2	90623.0	19480.4	29969.8
1 3/4	105726.9	22727.1	34964.8
2	120830.7	25973.8	39959.8

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M45

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	36592.6	8427.8	12965.9
5/8	40658.5	9364.3	14406.5
3/4	48790.2	11237.1	17287.9
7/8	56921.8	13110.0	20169.2
1	65053.5	14982.8	23050.5
1 1/8	73185.2	16855.7	25931.8
1 1/4	81316.9	18728.5	28813.1
1 3/8	89448.6	20601.4	31694.4
1 1/2	97580.3	22474.2	34575.7
1 3/4	113843.7	26219.9	40338.3
2	130107.1	29965.6	46100.9

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M48

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	43243.3	10623.5	16343.9
3/4	51891.9	12748.3	19612.7
7/8	60540.6	14873.0	22881.5
1	69189.2	16997.7	26150.3
1 1/8	77837.9	19122.4	29419.1
1 1/4	86486.6	21247.1	32687.8
1 3/8	95135.2	23371.8	35956.6
1 1/2	103783.9	25496.5	39225.4
1 3/4	121081.2	29745.9	45763.0
2	138378.5	33995.3	52300.5

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M52

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	47108.4	12537.5	19288.5
3/4	56530.1	15045.0	23146.2
7/8	65951.8	17552.5	27003.9
1	75373.5	20060.0	30861.6
1 1/8	84795.2	22567.5	34719.3
1 1/4	94216.9	25075.0	38577.0
1 3/8	103638.5	27582.5	42434.7
1 1/2	113060.2	30090.0	46292.4
1 3/4	131903.6	35105.1	54007.8
2	150747.0	40120.1	61723.2

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M56

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	60792.6	17424.0	26806.2
7/8	70924.7	20328.0	31273.9
1	81056.8	23232.0	35741.6
1 1/8	91188.9	26136.0	40209.3
1 1/4	101321.0	29040.0	44677.0
1 3/8	111453.1	31944.0	49144.7
1 1/2	121585.2	34848.0	53612.4
1 3/4	141849.4	40656.0	62547.8
2	162113.6	46464.1	71483.2

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M60

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	65430.8	20092.9	30912.2
7/8	76335.9	23441.7	36064.2
1	87241.0	26790.6	41216.2
1 1/8	98146.2	30139.4	46368.3
1 1/4	109051.3	33488.2	51520.3
1 3/8	119956.4	36837.0	56672.3
1 1/2	130861.6	40185.8	61824.4
1 3/4	152671.8	46883.5	72128.4
2	174482.1	53581.1	82432.5

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M64

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	69692.1	22828.3	35120.4
7/8	81307.5	26633.0	40973.8
1	92922.8	30437.7	46827.2
1 1/8	104538.2	34242.4	52680.6
1 1/4	116153.5	38047.1	58534.1
1 3/8	127768.9	41851.8	64387.5
1 1/2	139384.2	45656.6	70240.9
1 3/4	162614.9	53266.0	81947.7
2	185845.6	60875.4	93654.5

Aluminum 2024-T4/T351 ($F_{su} = 37.5$ ksi, $K = 0.20$)

Fastener Size = M68

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	74330.3	25869.3	39798.9
7/8	86718.7	30180.8	46432.0
1	99107.0	34492.4	53065.2
1 1/8	111495.4	38803.9	59698.3
1 1/4	123883.8	43115.5	66331.5
1 3/8	136272.2	47427.0	72964.6
1 1/2	148660.6	51738.6	79597.8
1 3/4	173437.3	60361.7	92864.1
2	198214.1	68984.7	106130.4

Aluminum 7075-T6/T651

$$F_{su} = 45.6 \text{ ksi}$$

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M1.8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	184.8	1.7	2.6

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	204.5	2.1	3.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M2.2

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	224.2	2.5	3.9

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M2.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	259.4	3.3	5.1

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M3

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/16	314.3	4.8	7.4

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M3.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	730.9	13.1	20.1

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M4

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	833.1	17.1	26.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M4.5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	943.1	21.7	33.4

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M5

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1052.8	26.9	41.4
3/16	1579.2	40.4	62.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M6

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1257.3	38.6	59.4
3/16	1885.9	57.9	89.1

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M7

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/8	1492.3	53.5	82.3
3/16	2238.4	80.2	123.4
1/4	2984.5	106.9	164.5

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M8

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2533.8	103.7	159.6
1/4	3378.4	138.3	212.8
5/16	4223.0	172.9	266.0

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M9

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	2886.3	133.0	204.5
1/4	3848.4	177.3	272.7
5/16	4810.5	221.6	340.9

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M10

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	3181.7	162.8	250.5
1/4	4242.2	217.1	334.0
5/16	5302.8	271.4	417.5
3/8	6363.4	325.7	501.1

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M11

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/16	3534.2	199.0	306.1
1/4	4712.2	265.3	408.1
5/16	5890.3	331.6	510.2
3/8	7068.4	397.9	612.2
7/16	8246.4	464.3	714.3

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M12

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	5105.6	313.6	482.4
5/16	6382.0	392.0	603.0
3/8	7658.4	470.4	723.6
7/16	8934.9	548.8	844.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M14

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	5969.5	427.7	658.1
5/16	7461.9	534.7	822.6
3/8	8954.2	641.6	987.1
7/16	10446.6	748.5	1151.6
1/2	11939.0	855.5	1316.1
9/16	13431.4	962.4	1480.6

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M16

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/4	6909.5	565.8	870.5
5/16	8636.9	707.3	1088.1
3/8	10364.3	848.7	1305.7
7/16	12091.6	990.2	1523.4
1/2	13819.0	1131.6	1741.0
9/16	15546.4	1273.1	1958.6
5/8	17273.8	1414.5	2176.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M18

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	9620.9	886.3	1363.6
3/8	11545.1	1063.6	1636.3
7/16	13469.3	1240.9	1909.0
1/2	15393.5	1418.1	2181.8
9/16	17317.7	1595.4	2454.5
5/8	19241.9	1772.7	2727.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M20

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	10795.9	1105.1	1700.1
3/8	12955.1	1326.1	2040.2
7/16	15114.3	1547.1	2380.2
1/2	17273.5	1768.2	2720.2
9/16	19432.7	1989.2	3060.3
5/8	21591.9	2210.2	3400.3
3/4	25910.3	2652.2	4080.4

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M22

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/16	11971.0	1347.9	2073.7
3/8	14365.1	1617.5	2488.4
7/16	16759.3	1887.1	2903.2
1/2	19153.5	2156.7	3317.9
9/16	21547.7	2426.2	3732.7
5/8	23941.9	2695.8	4147.4
3/4	28730.3	3235.0	4976.9
7/8	33518.7	3774.1	5806.4

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M24

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	15546.0	1909.6	2937.8
7/16	18137.0	2227.9	3427.5
1/2	20728.0	2546.1	3917.1
9/16	23319.0	2864.4	4406.7
5/8	25910.0	3182.7	4896.4
3/4	31092.0	3819.2	5875.7
7/8	36274.1	4455.7	6854.9
1	41456.1	5092.2	7834.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M27

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/8	17661.0	2440.6	3754.7
7/16	20604.5	2847.3	4380.5
1/2	23548.0	3254.1	5006.3
9/16	26491.5	3660.8	5632.1
5/8	29435.1	4067.6	6257.8
3/4	35322.1	4881.1	7509.4
7/8	41209.1	5694.6	8761.0
1	47096.1	6508.2	10012.6
1 1/8	52983.1	7321.7	11264.1

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M30

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	22805.6	3501.6	5387.1
1/2	26063.5	4001.9	6156.7
9/16	29321.4	4502.1	6926.3
5/8	32579.4	5002.3	7695.9
3/4	39095.2	6002.8	9235.1
7/8	45611.1	7003.3	10774.3
1	52127.0	8003.8	12313.5
1 1/8	58642.9	9004.2	13852.6
1 1/4	65158.7	10004.7	15391.8

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M33

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
7/16	25273.1	4268.6	6567.0
1/2	28883.5	4878.4	7505.2
9/16	32493.9	5488.2	8443.3
5/8	36104.4	6097.9	9381.5
3/4	43325.3	7317.5	11257.7
7/8	50546.1	8537.1	13134.0
1	57767.0	9756.7	15010.3
1 1/8	64987.9	10976.3	16886.6
1 1/4	72208.8	12195.9	18762.9
1 3/8	79429.6	13415.5	20639.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M36

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	31398.0	5785.1	8900.2
9/16	35322.8	6508.3	10012.8
5/8	39247.5	7231.4	11125.3
3/4	47097.0	8677.7	13350.3
7/8	54946.5	10124.0	15575.4
1	62796.0	11570.3	17800.5
1 1/8	70645.5	13016.6	20025.5
1 1/4	78495.0	14462.9	22250.6
1 3/8	86344.5	15909.2	24475.6
1 1/2	94194.1	17355.4	26700.7

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M39

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
1/2	34218.0	6830.1	10507.9
9/16	38495.3	7683.9	11821.4
5/8	42772.5	8537.7	13134.9
3/4	51327.0	10245.2	15761.8
7/8	59881.6	11952.7	18388.8
1	68436.1	13660.3	21015.8
1 1/8	76990.6	15367.8	23642.8
1 1/4	85545.1	17075.3	26269.7
1 3/8	94099.6	18782.9	28896.7
1 1/2	102654.1	20490.4	31523.7

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M42

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	41324.1	8883.1	13666.2
5/8	45915.7	9870.1	15184.7
3/4	55098.8	11844.1	18221.7
7/8	64281.9	13818.1	21258.6
1	73465.1	15792.1	24295.5
1 1/8	82648.2	17766.1	27332.5
1 1/4	91831.4	19740.1	30369.4
1 3/8	101014.5	21714.1	33406.4
1 1/2	110197.6	23688.1	36443.3
1 3/4	128563.9	27636.2	42517.2

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M45

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
9/16	44496.6	10248.2	15766.5
5/8	49440.7	11386.9	17518.4
3/4	59328.8	13664.3	21022.0
7/8	69217.0	15941.7	24525.7
1	79105.1	18219.1	28029.4
1 1/8	88993.2	20496.5	31533.0
1 1/4	98881.4	22773.9	35036.7
1 3/8	108769.5	25051.2	38540.4
1 1/2	118657.7	27328.6	42044.1
1 3/4	138433.9	31883.4	49051.4

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M48

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	52583.8	12918.2	19874.2
3/4	63100.6	15501.9	23849.0
7/8	73617.4	18085.5	27823.9
1	84134.1	20669.2	31798.7
1 1/8	94650.9	23252.8	35773.6
1 1/4	105167.7	25836.5	39748.4
1 3/8	115684.4	28420.1	43723.2
1 1/2	126201.2	31003.8	47698.1
1 3/4	147234.7	36171.0	55647.8
2	168268.3	41338.3	63597.5

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M52

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
5/8	57283.8	15245.6	23454.8
3/4	68740.6	18294.7	28145.8
7/8	80197.4	21343.9	32836.7
1	91654.2	24393.0	37527.7
1 1/8	103110.9	27442.1	42218.6
1 1/4	114567.7	30491.2	46909.6
1 3/8	126024.5	33540.4	51600.6
1 1/2	137481.2	36589.5	56291.5
1 3/4	160394.8	42687.7	65673.5
2	183308.3	48786.0	75055.4

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M56

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	73923.8	21187.6	32596.3
7/8	86244.4	24718.9	38029.0
1	98565.1	28250.1	43461.8
1 1/8	110885.7	31781.4	48894.5
1 1/4	123206.3	35312.7	54327.2
1 3/8	135527.0	38844.0	59759.9
1 1/2	147847.6	42375.2	65192.6
1 3/4	172488.9	49437.8	76058.1
2	197130.1	56500.3	86923.5

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M60

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	79563.8	24433.0	37589.2
7/8	92824.5	28505.2	43854.1
1	106085.1	32577.3	50118.9
1 1/8	119345.7	36649.5	56383.8
1 1/4	132606.4	40721.6	62648.7
1 3/8	145867.0	44793.8	68913.6
1 1/2	159127.7	48866.0	75178.4
1 3/4	185648.9	57010.3	87708.2
2	212170.2	65154.6	100237.9

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M64

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	84745.6	27759.2	42706.4
7/8	98869.9	32385.7	49824.2
1	112994.1	37012.3	56941.9
1 1/8	127118.4	41638.8	64059.7
1 1/4	141242.7	46265.3	71177.4
1 3/8	155366.9	50891.8	78295.1
1 1/2	169491.2	55518.4	85412.9
1 3/4	197739.7	64771.4	99648.4
2	225988.3	74024.5	113883.9

Aluminum 7075-T6/T651 ($F_{su} = 45.6$ ksi, $K = 0.20$)

Fastener Size = M68

[NOT TO BE USED FOR FLIGHT HARDWARE]

Thread Engagement (in)	Pullout Load (lb)	Assembly Torque (lb-in)	100 % Torque (lb-in)
3/4	90385.6	31457.0	48395.5
7/8	105449.9	36699.9	56461.4
1	120514.2	41942.7	64527.3
1 1/8	135578.4	47185.6	72593.2
1 1/4	150642.7	52428.4	80659.1
1 3/8	165707.0	57671.2	88725.0
1 1/2	180771.2	62914.1	96790.9
1 3/4	210899.8	73399.8	112922.7
2	241028.3	83885.5	129054.5